

Research on Auto Ethnography

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关键词

observational research 视角 儿童 成年人 心理 research project
ethnographic research life research own data more time certain times
research study research questions observational day researcher space
certain observations research process other systems Other days
design research own project auto ethnographic research

文字记录

说话人 1 00:00

So a bit of a long description, but I think one that helps us begin to position auto lithography not just as self reflection and just diary writing, but as a systematic process of using the self as a subject to better understand the world. So written here, we see that auto lithography is an approach to research and writing that seeks to describe and systematically analyze. I highlighted that here cuz we're not just writing down what we see and feel. We're making mean out of these observations, feelings. And that's what makes it research as opposed to, you know, just writing for the sake of writing. And we use these personal experiences that we've described and analyze to understand our cultural experience in specific context. So this is challenging what we would call the canonical or the traditional ways of conducting research, where research is conducted by an objective researcher on others. And we're looking outwards. This is the opposite of that. We are using the tenants of autobiography, so writing of self and ethnography to do and write auto ethnography. So this makes the process of auto ethnography build a process and the outcome of it being a process. But another way to describe it, if you're describing it to your friend to ask what you did today, you could say that you learn about the process of observing yourself a serving. Now that's a little bit meta, but that is the process of not just noticing and writing things down, but critically analyzing our own behavior, our own reaction in our own relationship to how we are walking through the world as a researcher and as a subject of that. Researcher, a subject of that research right there. So in fewer words, in the end, it's a qualitative method in which our own experience, so myself as a researcher, yourself as a researcher in your own project, is used as a primary data source. It's a sis systematic, right? So it's thoughtful. You're analyzing your self reflection and interpretations of cultural, social or system next phenomena. So that cultural social systemic is the ethno of auto ethnography and the origins that are in design research, but primarily coming from sociology and anthropology,

but now has become more info method when we're talking about design in future. Just practice and what we'll go over again and again. It's just something like a ghost.

说话人 1 02:51

Okay, I'm glad to post this here. Another thing, it's not just diary. Writing is a beautiful reflective practice that helps us understand ourselves better as people. But that's not just what this is, note taking about the world around us and our experience with it with a strategic purpose to inform how we interact with the subject for work event.

说话人 1 03:14

So what can this look like quite literally in practice? It can be reflected journally. Now this reflected journally can be around a specific event that you go to, a specific moment in time, can have specific prompts. It's not just writing in your diary about your day. You have a strategic intention for the journal that you're doing. This also kind of look like field notes or regular giants. Those kind of are synonymous with one another.

说话人 1 03:42

Self observation or recording. So setting certain times in which you are observing yourself during an activity, recording yourself either through audio diary or through video, retrospective analysis is the part in which you are reflecting back on what you collected and strategically going through to identify themes. It can come up in like artifact gathering and or also artifact creation, like collages way that it sometimes comes up for your making a series of artifacts over time. Here we just have an example from an auto ethnographic project of somebody's notes would probably only make sense to them, but were notes that were taken at a certain time of day over a certain period of time. And this is another, I guess we call artifact auto ethnography, where a person has plan to do certain observations in certain locations over periods of time. And this is the format they made for themselves to collect their own thoughts. So that's the difference between someone we could call maybe just self reflection for the se sake of self reflection, which might be more like diary writing versus, you know, strategic and planned collection of one of those awesome experiences. This is a project example that I feel like is turning the idea of self as up as subject up a notch of it. So we kind of just think about, okay, I'm going to make a change in my life that involves writing in my. I read every day after I do this activity or self recording myself.

说话人 1 05:22

This is a project in which the researcher was investigating time as a subject and to insert themselves into the research process. They live without clocks or any way to monitor what time it is for two weeks. So if we think about the method, here is a massive change in lifestyle. And through that method, they used a lot of different tools to capture how their experience changed given that intervention of completely depriving yourself of a sense of time. So in this way, the

person wants to internally ops observations of certain periods and rituals throughout the day. So we think the intervention is a deprivation of something. And the tools they use to capture the data is journaling, taking photos, audio recordings, and recordings of visuals.

说话人 1 06:19

There's a link to this project at the end of the slide. It's a really interesting way that they talk about electronic. I really recommend checking about. So there are some risks here. I think that in my own experience and what I've seen in projects before, sometimes auto with number, he can just sort of feel like the, like, what would we call that, like the sexy shortcut of like, oh, like I don't want other people like, oh, like, I don't have that much time.

说话人 1 06:48

I forgot to do my ethics. How can I collect data without doing those things? I'll just suppose myself as the subject. That's okay, right? It is a subject that's worth pursuing. However, it's not like a silver bullet solution because in the end, there is a risk of bias here, pretty massive risk of bias if you're your only research subjects. Some might call that like kind of quasi self indulgence of just researching yourself and you're being the only voice in there. And some people call it like naval music where you are just the being inward in a research project and forgetting that other systems, other people are part of this.

说话人 1 07:28

So to be able to hold both positions of participant and researcher, it does take a fair amount of work. It's not an easy shortcut in many ways that actually ends up adding a lot of additional work and trying to structure and scaffles the data that you collect to be both meaningful and personal, but also strategically meaningful to the project. The risk here also because self is subject and as we understand, given our positionality, given our relation. Relationship to the topics that we're working in, it can sometimes lead to just general generalizability. So just generalizing things that, oh, if I'm noticing this in myself, Delhi probably is chill if I'm noticing this in myself. So probably is too.

说话人 1 08:18

And we know that's not always the case. But once we get so deep into our own ethnographic, auto ethnographic research, these things sometimes feel like they become greater truth. So to mitigate that, we talk about triangulation a lot on this course, right? So if we have one data point, it's from auto ethnography. How are we connecting this data point with data from secondary research and data from group interviews to validate what is being asserted through the insights we gather? And additionally, if we talk about ethics, you don't have to do the same type of ethics forms when you're considering research on self.

说话人 1 08:56

It's still important to, etc. Is depriving myself of clocks for 2 weeks safe for me? Is, am I in the right mental safe for that? However, you do often reflect upon the people and experiences that you experience, that you are experiencing in your data collection. So how is me collecting information about my experience with my friend's problem? How am I protecting my friend and giving them just say what's being put in my project that makes sense? So if we think about it like this, if you can connect your individual experience that is being captured through your auto ethnographic work to a larger structure, a larger system, a larger problem beyond it, then it likely is gonna be grounded research that is serving your project. However, if it not, it might be a bit more scaffolded.

说话人 1 09:50

You might need to consider how can I bring in other voices, other sources, other forms of research to support what I'm saying here? Okay, so we are going to check this time. Okay. Okay, so we are going to have a worksheet, hopefully, I love for everybody. And this worksheet has two sides. So the first side is going to be about auto ethnography. And I'm gonna stop yapping for a little bit. And there are four questions here that I like to think of the general like temperature check, if this is a methodology that might fit for your project, it might not completely. And that is totally cool cuz this is a little bit of a niche approach, right? But I would like you guys to individually walk through these set of questions, give it a little bit of thought. If this isn't making sense to you or if you have questions about a caller and maybe we can like also as a group talk about it a little bit more. But let's get started with this. We'll go through this for about 10 minutes and then share with a partner for about five. As I'm testing this out, does anybody have any specific question? She got started.

说话人 2 11:18

Alright, sure. And then I might just let you guys actually let me know if it's not enough.

说话人 2 11:49

But I think so.

说话人 3 12:01

Finding a person.

说话人 3 12:42

Consider you.

说话人 3 12:51

Canadian.

说话人 1 13:09

Discreetly and answering research questions, I think that even has like a further the work concept for how.

说话人 1 14:20

Yes.

说话人 2 14:21

And I would say it's more of what, how are you cases and then how are you analyzes, what do you have there?

说话人 1 14:34

So if you just when do the process didn't sort of like dramatically understand what was something else to you and you just have like schedules without any real analysis.

说话人 2 14:49

I would say that probably a strategy.

说话人 1 14:54

However, yeah, as a part of your journalist, you have an angle in which you have analyze the work that you've done. And so we only have a certain observations that we involve in additional voices.

说话人 2 15:21

I couldn't have like voice requested me.

说话人 3 15:26

Sure.

说话人 1 15:28

I think that what we've done, that's gonna do not enough so on as well.

说话人 2 15:32

And I think that you should like amplify what you've been saying, provide more academic.

说话人 1 15:45

So maybe we filling out that considering from like an analysis perspective. So we don't work on it. Yeah, pretty much naturally collected. I don't mind confused unless I think you're done process. It's the step. So like we give the independent. Yeah, how's it going on? Any questions now? Do we have enough paper for everybody?

说话人 3 16:30

Yeah, yeah, yeah, I see.

说话人 1 16:35

I know that there's been this already. Great.

说话人 2 16:39

Cuz you can do in your interview functions today, solution for the eye.

说话人 1 16:54

Oh yeah, wait. So make sure you're doing the auto ethnography set. There you go. So there's two sides. One has EATL.

说话人 2 17:02

You do another side many times.

说话人 1 17:19

Maybe it's not always like it's inevitable, right? Oftentimes it's like probably confidence in your own experience with like other forms of data that ensure that the outcome request is just informed by those single activations. Yeah, so it's, I think, mostly complementary or also sort of like structural video, like how am I respond into these things? And then more video cases like, okay, substructure consultation, are you thinking about this?

说话人 1 18:00

Or like you're in a request and when you're doing this in the industry, you're not always like given the same quantity of our time and energy to, etc.

说话人 3 18:11

用民族志的方式研究别人学习民族志。很好玩。

说话人 3 18:33

如果不用笔的话，先把笔这个有需要的，要不就现在用这个笔很粗的。 I think.

说话人 1 18:53

These are all of the questions, see what business takeouts.

说话人 3 19:00

Embedding? Embedding 是什么意思？嵌入。嵌入是什么意思？是我在这个系统里吗？就你自己有没有这个问题？那就是他。yes。

说话人 1 19:23

yes。

说话人 3 19:26

儿童心理健康教育。

说话人 2 19:28

你是儿童。对啊。

说话人 3 19:31

我是儿童。所以我肯定在这个系统里面。不是，主要我现在不是，但我曾经当。

说话人 2 19:40

那你现在在讨论的不是现在的视角。对。对的。对，不同的。那你现在还在。

说话人 3 19:50

作为一个成年人讨论儿童的视角，作为一个，那我就是作为一个成年人的视角。

说话人 2 19:57

那你这个。

说话人 3 20:00

有新视角。

说话人 3 20:09

8C 的水还是如此的冰凉，让我脑袋非常清醒。

说话人 3 20:19

是的，你连第一问题都没回答。回答了。yes or no。不是，我们的题怎么不一样？你刚才在看什么？为什么？是这个，你是对的。对呀。你是对的。我是对的。Great.

说话人 2 20:43

Okay, y'all.

说话人 1 20:44

I understand that probably didn't get to the end of that paper and that's truly okay. This is something that is, you say homework, you sit with, you recycle it, you don't think about it, whatever. But before we move on to the next method in filling this list of questions out, regardless of if you're planning on using this method or not. Does anybody wanna share what

came up, what they rode, what the experience was like that I can listen to the rain all day, guys, don't worry about me. Thank you.

说话人 4 21:37

Thank you. Yeah, and so I've already kind of started my mapping graphic study and things around a hard time. So that's really bad. I think the whole way drive me kind of distributing myself and fear of being biased or like you say, negative agent. I think maybe we would be quite impressions to do this structure also economically alongside the other participants and kind of doing a magnetic coating and separate myself, but maybe you visit the here's me as more informed on the subject versus the participants and maybe there's some video game I expect comparison.

说话人 1 22:26

Yeah, it's a really good question, especially. Great. Yeah, nice to meet you, Shannon. So I really like what you said around specifically around when you're conducting ethnographic research, you but I'm be so careful about like, where does my experience and understanding and voice stop and where does what I'm observing and what I want to be taken in stage. Sir, sponsor, how are you been doing that? Like up till now, I guess, how are you guys mindful about consideration?

说话人 4 22:53

So I just can be like a previous and study family with the results yet. So I'm going, yeah, so like I said, I really considered myself as being part of that separate information.

说话人 1 23:13

Yeah, that's good to hear. And I think.

说话人 1 23:23

Especially when I go or sometimes. We're gonna move on to day in the life before we get to stay like, does anybody have experience doing data like on observations or observational studies? Yeah, it's not you want to share, you give me a big old map. I had to ask. Okay.

说话人 5 23:49

Oh yeah, I did it like last year. And it's a researcher about how we use our like phones, like social medias or like access you choose or like these apps. And while like we're on our way to work and while we're having like a lunch break and what kind of apps we use like after we go to work or like before bed and like the first app you use when you wake up. And I did like, so such a research for in bargain day for like what kind of apps would you think like some time in the in my day.

说话人 1 24:31

And when you were observing people, were you monitoring your screens or were you question and person?

说话人 5 24:37

Actually, I like, it's like, did you avoid a essay test like this? You actually, so I just ask them, like to ask them know what kind of apps they do because I think like have cameras or like sunset access to monitor their lives.

说话人 1 24:54

Yeah, yeah, that's actually really fine. We'll hold on to that as we move forward about like when asking could potentially lead to like unproductive results. When does like watching, sorry. Which I think that's when we're talking about like, , that's gonna be a fixed ethics application, but also in a world in which we have maybe more time, energy, money, resources, how does observing and sometimes like tend to take precedent as a preferred method? Thank you so much for sharing.

说话人 1 25:24

Okay, so we'll talk about David. Like I think most of you should be somewhat at least familiar with just what observational research is. It kind of does what it says. On the hand, you're watching something for someone. It's an ethnically, ethnographically rich method in which you as a researcher are kind of. Following something that can be quite literally following them, like following down the street or that can be following along with their screen following and along with the moment in their day. And this can be a person, community or practice. And it's through a structured window of time.

说话人 1 25:56

So similar to when we were talking about auto ethnography, there's structure there, right? There's structure about how we're capturing data about ourselves. When we're capturing data about ourself, why are we capturing data about ourself? Similarly, we have structure. When we are following or observing somebody. It's not just free brawl. And if there's the structure, obviously, specifically around that fix. And the purpose of this, and the most part is to capture rhythms of one's life, transitional periods, frictions, which will come up a lot when you're talking about like CX work like you were doing and workarounds that are occurring in real time, not when they're recall. So in the example provided, that's what's catching the gap between what stated behavior and what's actually happening, right? So what one states can be influenced by many things, right? Comfortability, but the interviewer, if they're, you know, like confident in the answer, how the question was worded. Like if they wants to be honest about what they actually feel, a myriad of things.

说话人 1 27:10

Even though there are influences when it comes to being observed, we can talk about in a second it, the process of observation and following the day, the life of somebody does lessen the gap between stated and actual behavior. And the temporal structure or the way that the time in which you're observing someone is quite important because the passage of time and the experience of being with somebody over a certain period of time provides a little bit more depth and lived experience than a survey that tends to Latin responses. There's much less time, energy, space, quite literally, to provide like the human texture of one's experience.

说话人 1 27:53

So as we know, systemic forces and conditions don't always announce themselves. If I was giving a free response on a survey, my response would be informed by the words on that survey, the time that I did it, who sent it to me, how I received it. But deeper systemic conditions can often be identified by observing somebody in their natural space, in the natural rhythms of their life. Things that emerge might not be things that the observer or the researcher would think to even ask, and also might surface things that the observi or the participant might not even think to offer, might not even think is relevant to the project, not because they're ignorant. Other days, like on the contrary, that the rhythms of their life are so deeply ingrained and they're on the experience. So they might not bring that to the environment of the researcher space cuz it might not feel relevant.

说话人 1 28:58

As we know, when we're talking about behavior change, routines and ritual in encodes value and encodes meaning and encodes the systemic workings of one's life. There are workaround specifically when we're talking about CX observation, when you're observing somebody using, let's say a digital app or computer space that people who are the developing the app might not understand. There are workarounds when using a tool or a system or a process that aren't easily identifiable unless you witness somebody actually applying those workgrounds.

说话人 1 29:33

A time in which observational studies tend to happen, tends to provide live data also is through transition states where stress can occur. Transition states can translate really broadly, right? That might be from being asleep to waking up. That's a transition state. Might be from getting on the bus in half me and off the bus in Kensington. And we're looking specifically as designers at places where the design environment that we design for is meeting the lived environment that we live in. From a future's perspective, it provides a really rich and textured view at that early horizon 1 thinking around the observed presence and that observed present is what we can push our speculative scenarios again. So it's grounded in human reality, human behavior and what is happening quite literally in a day versus like today.

说话人 1 30:32

So what does this look like in practice? This can look like in person shadowing where you are kind of quietly following someone amongst their daily life. It can look like an out. Out of timestamp field notes, photo diaries where you're taking images of person's day, video recordings like was mentioned. Yeah, it sounds nuts to have set up a bunch of cameras in people's houses, but maybe not for this project, but for future projects that can like literally happen for large scale research projects, obviously takes a lot of ethical reviews, retrospective interviews that you gather the interview material while you're observing them. I'm not conducting an interview while I'm observing them, but I'm getting my own data that I can then ask them about later. It's the data and the questions that I'm getting are informing the research to be more brand and next review. So these are some questions that we consider. I won't go through them all. You can also read them.

说话人 1 31:31

What do specific routines look like or do they feel like? How do they change in certain factors? How do products or services fit into a business life? So this is one example, not screen idea about the end, but the I thought is like a particularly interesting observational study as it looked at like a very intimate time in a family's day. So this was an observational study conducted with new parents and young babies. And the researchers observed like the barriers around feeding our, of like getting ready to feed your child, feeding your child and put it into the way and notice all of the pain points that the parents were experiencing around for whistle and frictions they were experiencing around the packaging of the food that they were providing for their child. After a study, a lot of different families, they collated their findings and provided their client Funnel Organics with a new packaging concept for their future products that would more greatly meet the needs and meet those friction that the parents are identified.

说话人 1 32:49

So there are risks here. The first one that I think that like is kind of a gut feeling on if we're subject to these interviews is called the hawker, in fact. And that is your behavior changing because you know someone's watching. Like I even thinking, I mean, when I was making these slides, I was thinking about like, what would I like feel comfortable with someone observing? And even if they we're observing that, I'm guessing my behavior would change. That's a little bit assumed in observational research, but it's still important enough method to practice and keep that in mind.

说话人 1 33:26

Additionally, similar to auto known be over generalization, right? Just because you maybe did this with one person and for this one person, something atypical happened in their day, and you might see that as concrete data, but it's not, right? Our lives are complex. Things happen. There is a risk of overgeneralizing a unique data point would be fine just cuz someone was having a

weird day. A big one here is around access and power. Similar to one here. Like I think I could also be around. You can put videos in their house because it is a really like intimate process, even if you're just following someone around the. And it's intimate to be perceived, especially knowing that what is being perceived is going to be like analyzed and decisions will be made, right? So many power dynamics and considering how your research can be as non extracted as possible. So in some businesses, you know, people can be paid for these things in academia, not so much. So how can there be relationship with more reciprocity where they're getting their results back immediately, where they're being listed in the in like the co authorship. How can you be welcoming them into the process considering your positionality, who you are is influencing what you notice, right? So if I am conducting this research study and I watch the same exact day and CJ does the same thing, and you watch the same exact day, we will probably have different sites because it's different people and who we are is informing in some ways what we notice and then also a lack of structure. So we talked about that in automatography, you just need to provide barriers from why you're looking at, what you're looking at, how you're doing it, etc. Okay, so applying it to your final major project where you're in a research project rather. So now we have side two of our sheet and I'm gonna welcome you guys just to go through the next three part. So hey, we don't get them to them all and these questions are prompting you to consider. Consider how observational day in the life research could potentially be welcomed into your project. Once again, I'm honest expecting you guys to be able to video take one of your friends in their house for like two weeks. But there are other small ways that you conduct up observational research, like we discussed, like shadowing something tycoon and areas like short video recordings of certain.

说话人 2 36:08

Yes.

说话人 1 36:09

So let's spend the next 10 minutes walking through these questions. Just considering if in a world you decided to use this method, who might you follow? How might you do it? When might you do it? Like what we know of one, say, what activity is worth you certain and how would you keep your participants and yourself and keep it safe and consider this interaction. Yeah, so this would require an ethics approval because you are using a person as a subject. It would just in a combination, they need to know.

说话人 1 37:03

Okay, that's actually a clarifying question. So this is research. If you're conducting a day in the life of somebody, that's different than just sitting at a mall and watching people go by. You are like specifically working with, like you are serving me and you reach out to me and say like, hey, I know you're a surfer and I like watch you go about your day regarding the surf. So think about

sort of like the exact person or entity. So this is different than just watching a mass group of people who you don't have permission to, users.

说话人 1 38:10

What?

说话人 3 39:56

China. Right. Yes.

说话人 1 40:41

It sounds abilities.

说话人 1 41:20

Okay, would anybody like to share what they were done? I'm very curious to see what you guys think about this method in addition to your projects.

说话人 1 41:43

I love the silence. Oh, oh, we two beginners. Okay, here. Okay. Okay. I am planning to use like a. Yeah, yeah, but more like the cultural profit, like not the whole day I'm working on like social media is like among the internet for UK. So I was thinking of getting them to share the screen recording for the TikTok or like social media games and I wanted to analyze it. Yeah, so I was thinking like the time frame for me, like I know it's a lot. So about three days minimum each, right? So we don't change everything as much. And perhaps I don't follow up into me to ask them again about, oh, what they saw. Yeah, something like that. And are you study having done the analyst today, right? Yeah, I like that that like I feel like immediately handmade that I feel like that would increase the comfort level as well. They're the ones that are analyzing their own information. And also that's like a nice way for, you know, to support lens strap. Keeping updated on that one. I'm very curious to see what services. I think I said, yeah, session with you.

说话人 2 43:02

Like I remember like some sort of consultants.

说话人 1 43:05

So yeah, let me give you updated. I'm fabulous. I'm like ready.

说话人 2 43:11

My name is so exciting.

说话人 1 43:13

Hey, that's okay. That's okay.

说话人 2 43:16

I don't know reasons like who really has that for situation. If I were to use it, I might, for example, was a big teacher, could you observe a lesson plan, details, AI, how they use it?

说话人 2 43:45

This would help with just an interview if they have any hesitation, concerns, what platforms might look like, how much information. Okay, that's a little bit of shame. But if you think about it, like, oh, I don't use a whole thing, just use the brainstorm people. Obviously, the account, the whole thing, see what comes to put in account that goes into the lesson plan.

说话人 1 44:26

That's juicy, especially when we're thinking about like, you know, it's a little bit cliché, but you think it's not like that. The building side of things, they'll like literally see how much an educator is needing to do it and put in a certain amount of time. And under sort of like from your perspective, the understanding like the opposite side. And I guess that, yeah, that's very interesting. And I guess it would be like anonymous. So they, I love feeling more comfortable.

说话人 2 44:55

So important during lesson to the afternoon. Really cool. Yeah, cool.

说话人 1 45:06

Any other ideas or inspiration from what we've heard from our two to share so far? I'm feeling inspired.

说话人 2 45:11

I'm not doing research right now.

说话人 1 45:16

Yeah, is it possible?

说话人 2 45:20

So like my thought on how to apply it is to do an auto ethnographic day in the life of myself in 2050.

说话人 1 45:32

I love that. Is that possible? Hell yeah, that's possible. And that's punk too. I love that. Okay. Okay. So it's more as output than input, right? Like it depends on Horizon 3 and horizon 1. Yeah, so it can be a 5 months ways, right? So if we're talking about like how this would be deeply horizon 1, how are usage before an understanding of the present. Also when we're talking about

how are we translating what the future could, when should maybe delay this? Also be an output of like, how are we understanding the day. So it's like in this future that you imagine, how is it they know like we're not actually providing like a really dish texture to and imagines future. Keep me updated.

说话人 1 46:19

Yeah, yeah, exactly. Oh, cuz I'm so creative. Okay, I wanted to give you some questions. The first thing, cuz I know not all of your project first are human centered. Some of them are animal centered, planet centered, techno centered. What would it mean for like using more than human perspectives in relation to like daily life method? How could your project is about animal interaction with natural right? Sorry, I was so confident that somebody this somebody on this course is and I was thinking about like, how could they follow around a very specific cat all day and on the day in the life of this like neighborhood, is that person in the room right now who's doing animal interaction with London Europe? Okay. I've been waiting for that and it did it clock big time.

说话人 1 47:11

Okay, so considering other of non human actors in a project and this project or in future projects, how could you follow the day in a life of a certain tree in a park and how the outside world interacts with them. How can you follow animal? How can you follow a system or an idea? Just generally, I think you'll leave this room being like, they don't talk about a lot of risks. There are risks and spend the benefits to really outweigh them as long as you're careful. And I know that you guys are careful when researchers, and I think that this question's kind of thing like ask an answer. How can we apply this like daylight method to imagine futures? And it might just turn into output, it might turn into workshop, it might turn into a skit or something. These things can live the present as technical ethno graphic methods that we use and we write about in our academic way, they also can be extrapolated into tools of communication. You can combine them, machine together. And that's all I have for you guys today. Thank you, cuz.